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ORIGINAL RESEARCH

BioCrypt-DNA: A DNA-inspired biocryptographic framework for secure cloud computing

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Abstract
With its unique functionality, cost-effectiveness, and ease of use, cloud computing (CC) has transformed the modern world. It involves self-service provisioning of computing resources, such as processing and storage, via internet-accessible data centres. New and innovative technologies make it easier and cheaper to utilise computer resources. This paper investigates CC and biocryptography to improve data security. It suggests a new methodology, BioCrypt-DNA-based Cloud Security, that leverages DNA's distinctive features to store and encrypt data in the cloud. The research focuses on the potential of DNA cryptography to address the challenges modern encryption methods face, especially as cyberattacks become more prevalent. The paper demonstrates how DNA encoding, combined with cloud storage, can create a scalable, efficient, and secure solution for preserving sensitive information in the cloud. The proposed framework leverages DNA-inspired symbolic encoding within a fully digital cloud environment, ensuring computational scalability and cost-efficiency without relying on physical DNA synthesis or laboratory infrastructure. Furthermore, this study examines the feasibility, challenges, and implications of DNA-based encryption in real-world cloud environments.

Keywords Cloud Computing · Data Security · DNA · Encryption · Decryption · Cryptography

1 Introduction

With incredible cost efficiency, self-service, and ease of use, CC has changed how individuals store, process, and manage data. CC is the newest paradigm in information technology due to its ability to provide on-demand access to data processing and storage at available internet data centres. It enhances the efficiency with which organisational and economic computing resources are utilised, helping promote hyper-growth innovation across industries. Nevertheless, the need for robust security measures is increasing as reliance on cloud storage grows (Karimov and Olimov 2021). The growth of generated data and more sophisticated cyber threats has epitomised the failure of standard cryptographic techniques and highlighted the need for more secure encryption methods. Conceptually, the fundamental unit of life, DNA (Deoxyribonucleic Acid), offers extensive opportunities to safeguard information using a CC framework. DNA possesses distinct properties that make it suitable for storing and securing information (Suganya and Sasipraba 2023). Given that it can store vast amounts of data in highly compressed, instantaneous formats and that it has inherent cryptographic properties, DNA undoubtedly emerges as a powerful medium for protecting sensitive data (Qiqieh et al. 2025).

The idea of using DNA as a storage medium is not new, but the fact that it can be used as a cryptographic tool radically changes the story of information security. Scientists have begun exploring the limitations of the given paradigm and

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building a framework better suited to cloud systems. This is achieved by integrating DNA storage with cryptography to provide greater security than traditional approaches. Therefore, embedded encryption systems based on DNA. For decades, symmetric and asymmetric encryption have effectively provided data security. The problem, however, lies in the increasing volume and complexity of cyber threats. Modern brute-force, side-channel, and quantum-computing attacks have shown that traditional encryption algorithms can be easily overcome. This necessitates novel approaches to counter sophisticated cyber threats while ensuring optimal cloud data security. DNA cryptography uses principles of molecular biology and DNA's characteristics in previously unimagined ways. DNA cryptography encodes data into DNA strands, creating an entirely unbreakable form of security when combined with biological encryption. As a result, it serves as a viable solution to these problems (Jero and Misbha 2025). The connection between DNA encryption and CC offers new opportunities to improve information protection. The enormous storage capacity and robustness to perform encryption with no possibility of decryption further reinforce its position as the principal encryption strategy for massive, cloud-hosted datasets. Furthermore, the chaos and variety in the order of DNA sequences pit the sophistication level for unauthorised users against highly sophisticated systems that are difficult to manipulate (Singh et al. 2024). This new method overcomes the weaknesses of traditional cryptographic methods and describes a world in which biological and computational practices coexist to build secure systems.

The paper discusses opportunities to leverage DNA encryption to address information security issues in the cloud. The most important thing is to research the theoretical foundations of DNA cryptosystems and their use in a cloud system (Liu et al. 2024). Once the reasoning, methods, and consequences of implementing such DNA-based encryption are studied, I hope to contribute to the existing literature on genetic cryptography and the protection of information in the cloud. The research aims to explain the application and viability of encryption technology and its DNA dependencies, focusing on the magnitude of impact on data security in the cloud. The change in how digital resources are protected, accompanied by the utilisation of DNA in encryption, is the two phenomena at the core of this problem. DNA cryptography is a unique encryption method involving biomolecular structures. It employs the vast storage capacity of DNA, which outshines any known storage device in both size and efficiency. DNA-based encryption is further aided by CC infrastructure with powerful processing capabilities. The provided infrastructure is highly efficient in safeguarding important information, making it well-suited for confidential data protection. As a branch of molecular biology, DNA cryptography originates from the flow of genetic information from DNA through RNA to proteins. DNA is a double-stranded helix made of nucleotides consisting of four bases, which are adenine (A), guanine (G), cytosine (C), and thymine (T). These bases form complementary pairs: A with T and G with C. Bases can be used to encode and decode information. It is possible to construct extremely secure encryption schemes that withstand conventional cryptographic attacks by mapping digital information to DNA sequences. Encoding messages onto DNA strands for encryption introduces a level of complexity and randomness that is hard to duplicate. DNA is therefore a perfect candidate for safeguarding data stored in the cloud. Moreover, DNA offers numerous advantages beyond its cryptographic potential. With its high storage capacity, DNA is a very good choice for long-term data archiving. Moreover, DNA is most resilient and can last thousands of years under favorable environments, and thus it is very durable, and therefore the preserved information has a long shelf life. By integrating with CC, these properties, along with DNA's versatility, provide a defensive advantage for securely storing large amounts of data and efficiently accessing it. The approach to CC and DNA-based encryption solves many problems that existing cryptographic techniques face. For example, protecting information using DNA sequence encryption requires a level of complexity and randomness that is very difficult for a user to access, thereby preventing side-channel and brute-force attacks (Yuvarani and Mahaveerakannan 2024). Moreover, the data is encrypted, so the correct key is mandatory to decrypt the information. This characteristic greatly increases the protection of the information. It is reasonable to suggest that side channels for achieving sensitive data in the cloud may be more difficult for attackers to exploit. This paper presents a new framework called Biocrypt-DNA, which is a DNA-based cloud security framework that combines DNA encryption and cloud storage for data security. Traditional DNA encryption is combined with standard cryptographic algorithms to develop a solution that uses DNA for sensitive data protection in cloud environments. This system proposes a reasonably modern and scalable encryption technique that exploits modern DNA cyber threats. It works through several steps, from security access management to user data decryption and retrieval for secure information. Fragmented, sensitive DNA-encoded data is stored in the cloud. Users' sensitive data integrity and security are further strengthened at each step, while preventing unauthorised access. The main contribution of the paper is given as follows:

- We introduce a novel DNA-based encryption methodology for cloud data security, combining DNA encoding with traditional cryptographic algorithms.

- We utilize DNA's inherent cryptographic properties to improve data confidentiality, integrity, and availability in cloud storage.
- We offer a scalable solution for secure data storage by fragmenting and distributing DNA-encoded data across multiple cloud servers.
- We implement secure access control and key management systems to prevent unauthorised access.
- We test the efficiency of the technique with respect to the time taken for encryption/decryption, key space analysis, and attack resistance.
- We review modern attempts at DNA-based cryptography and its new applications to the raging problems of cloud security and set the ground for enhanced studies.

The rest of this paper is structured as follows. Section 2 reviews previously published studies on DNA cryptography and cloud security. Section 3 covers the aspects of DNA computing and its relationship with CC. Section 4 focuses on the problems and security issues of CC, emphasising the need to develop new safeguards against unauthorised access. In detail, Sect. 5 proposes the BioCrypt-DNA methodology, which explains the process of DNA encryption and its applications in CC. Section 6 presents the results of evaluating the proposed method: encryption, decryption, and analysis of the key space, including estimation of the technique's robustness against various attacks. Section 7 is responsible for carrying out system security analysis that involves the system's ability to resist integrity and confidentiality threats. Lastly, Sect. 8 concludes the paper by reiterating the key points and outlining how DNA encryption could affect the security of future CC solutions.

2 Literature review

Recent progress in DNA computing and the use of DNA to store data have raised new questions about the secure representation of information. DNA-inspired cryptography uses the four-base alphabet (A, T, C, G) and, according to this format, complex coding protocols are built to add complexity and diversity to the key space. In some studies, DNA-based encryption, a hybrid bio-cryptographic and sequence-based data masking, has been investigated to provide resistance to brute-force and structural attacks. The theory of storing biological DNA at super-high density is promising, but the cost of synthesis and latency limit its practical application. Therefore, the concept of digitally simulated DNA-inspired encoding has become an effective option toward improving the security of structures in more traditional computing systems. Cryptographic protection with structural obfuscation in the context of CC, where information must be spread across multiple nodes and is vulnerable to advanced cyberattacks, provides an added layer of defence on top of standard encryption. Thus, DNA-based encoding solutions are considered today as one of the potential complementary measures to enhance confidentiality, fragmentation-based resilience, and distributed data protection. The analysis of previous literature indicates that more attention is being paid to research on DNA cryptography, which has the potential to address issues in cloud data security. Some work has been done to investigate the basics of DNA cryptography, particularly the encoding methods and their use in CC (Namasudra et al. 2020; Verma and Jain 2015). Other scholars have researched cloud security problems, highlighting gaps stemming from conventional cryptographic techniques and advocating for DNA-based encryption (Shanmugasundaram et al. 2017; Elminaam et al. 2022). Some of them focused on concepts of DNA data encodings and the prospects for compact data storage (Wang and Song 2017, Hamed et al. 2016). In contrast, others concentrate on DNA-based encryption techniques for CC, emphasising privacy and scalability (Kumar 2021; Namasudra et al. 2018). However, the sharpness of some of the theories is diminished in practice because of the absence of the desired details, scalability analysis, and overall computation complexity and cost analysis (Elminaam et al. 2022; Kairi and Bhadra 2023). Nevertheless, the work prepared so far leads to an inescapable conclusion: the promise of DNA in solving cloud security problems is more than wishful thinking. This will be possible through DNA's ability to store data at extremely high densities and the prospect of developing cloud-based systems with virtually unbreakable encryption.

Table 1 presents my findings on how DNA cryptography is being applied to enhance cloud security. From an initial understanding of the principles of DNA cryptography and encoding methods to examinations of security threats on the cloud and DNA-enabled solutions, they encompass a wide range. Some studies focus on DNA-based secure cloud storage, privacy "scooping," and the scalability of DNA cryptography. As with other academic work, there is room for improvement in practical implementation, scalability testing, and verifying assumptions regarding computational efficiency and cost-effectiveness. The research collectively contributes to the growing body of knowledge on DNA cryptography and its potential to secure cloud-based systems. In contrast to earlier methods in DNA-based systems, which mostly consider the feasibility of biological storage, the proposed BioCrypt-DNA architecture integrates digitally simulated DNA coding with a distributed cloud system to enhance structural security without sacrificing computational feasibility.

Table 1 Comparisons among existing studies

Study	Description	Limitations	Contributions
Namasudra et al. 2020)	Explored fundamental principles, encoding techniques, and potential applications of DNA cryptography	Limited practical implementation and scalability testing	Introduced DNA cryptography principles and encoding techniques for CC security
Verma and Jain 2015)	Surveyed cloud security challenges and proposed DNA cryptography as a potential solution for enhanced information security in cloud storage	Lacked concrete solutions or implementation details	Highlighted the need for innovative security approaches and proposed DNA cryptography for cloud storage
Shanmugasundaram et al. 2017)	Reviewed recent DNA data storage technology advancements, emphasising their potential for compact data storage	Primarily focused on storage, with limited discussion of cryptographic applications	Summarised advancements in DNA storage, laying the groundwork for cryptographic applications in the cloud
Elminaam et al. 2022)	Examined various cybersecurity aspects in CC, highlighting limitations of traditional cryptography and suggesting novel approaches like DNA cryptography	Lacked a detailed exploration of DNA cryptography techniques and feasibility	Identified limitations of traditional methods and advocated for novel approaches like DNA cryptography
Wang and Song 2017)	Investigated a DNA cryptography-based approach for secure cloud storage, emphasising its enhanced data security and confidentiality potential	Limited experimental validation and scalability analysis	Proposed a DNA cryptography approach for secure cloud storage, showcasing its potential for enhanced security
Hamed et al. 2016)	Explored DNA encryption procedures for privacy-preserving information storage in cloud-based systems	Focused on privacy preservation but lacked a detailed performance evaluation	Investigated DNA encryption for privacy-preserving cloud storage
Kumar 2021)	Analysed DNA encryption algorithms for enhanced cloud data security and potential applications in secure cloud storage	There is limited discussion on integrating DNA cryptography with existing cloud security frameworks	Analysed DNA encryption algorithms and their application in cloud data security
Namasudra et al. 2018)	Examined the scalability of DNA cryptography for cloud-based applications and its potential for flexible and secure data storage	Did not address the computational complexity and cost of large-scale implementation	Explored the scalability of DNA cryptography for cloud applications
Elminaam et al. 2022)	Compared encryption techniques for secure DNA information storage in cloud environments	Limited focus on real-world deployment challenges and cost-effectiveness	Compared encryption techniques for DNA-based storage in secure cloud environments

3 DNA computing

Biocomputing is a current trend that leverages the computational complexity of bimolecular concepts alongside traditional cryptographic methods to develop secure algorithms (Kairi and Bhadra 2023). The application of bimolecular encryption techniques has spawned a novel branch of cryptography known as DNA cryptography, whose principles are primarily centred on DNA (Namasudra et al. 2018) and the central dogma of molecular biology. DNA, or Deoxyribonucleic acid, constitutes a double-stranded helix composed of nucleotides, where each nucleotide consists of one of four bases: A (adenine), G (guanine), C (cytosine), and T (thymine) (Kairi and Bhadra 2023; Kairi et al. 2023).

3.1 DNA and cloud computing

An upcoming NA data storage technique involves storing digital information in DNA molecules. Since DNA is a highly durable, dense storage medium, it may meet the growing demand for effective long-term data storage (Benyahia et al. 2025; Yuvarani and Mahaveerakannan 2024a). In contrast, CC uses an Internet-hosted network of distant computers for data processing, management, and storage. Physical storage of DNA can be seen as having extremely high potential density and longevity for archiving, though current DNA manufacturing and sequencing methods are characterised by significant latency, high cost, and reliance on labs. These are restrictions on their use in real-time cloud transactions and high-frequency data processing. Accordingly, the proposed BioCrypt-DNA framework does not use biochemical

DNA synthesis in its operation. Rather, it provides a digital DNA abstraction model that runs entirely on a computational cloud infrastructure.

The methods listed below can link DNA data storage with CC.

- *Transfer and Accessibility of Data* CC platforms can efficiently transfer and store data produced by DNA sequencing in labs. Researchers may upload vast volumes of DNA sequencing data to cloud storage, from which other users or computational processes can access it for analysis and interpretation.
- *Computational Data Analysis* CC provides the computational power needed to process and analyse the large volumes of data generated by DNA sequencing.
- *Storage and Management of DNA Data* While storing DNA data in actual DNA strands is still under development, cloud-based databases can store, organise, and manage metadata, annotations, and other information linked to DNA sequences. This makes the indexing and retrieval of DNA data efficient.
- *Parallel Processing and Scalability* As the volume of DNA data grows, it is possible to distribute and parallelise DNA data processing operations across a cloud infrastructure, enabling faster analysis and greater scalability. A cloud system can dynamically provision resources in response to fluctuating demand when processing DNA-related activities.
- *Integration of DNA Data Storage Concepts* As DNA data storage technologies develop, cloud service providers (CSPs) may adopt these concepts, enabling data stored and processed in DNA molecules to be easily accessed and manipulated. This may involve DNA-encoded data, tools, or APIs.
- *Data Privacy and Security* To address data security and privacy laws, CC platforms may implement advanced security systems to protect the privacy and confidentiality of DNA data.

3.2 Biochemical-to-digital interface in Biocrypt

The suggested BioCrypt-DNA framework does not rely on the actual synthesis of physical DNA or laboratory sequencing in the encryption and decryption processes. Rather, the system relies on the digital representation of DNA sequences, with the nucleotides A, T, C, and G serving as symbolic representations of binary data. Traditional DNA data storage systems subject biochemical processes such as DNA strand synthesis, amplification, and sequencing to introduce latency, increased infrastructure costs, and reliance on laboratory infrastructure. But BioCryp-DNA is fully a computational system, with DNA encoding and decoding implemented in software via mapping functions.

The bio-chemical-to-digital interface in BioCrypt, hence, comprises:

- Classical algorithm (AES/RSA) digital encryption,
- Encoding into the symbolic DNA by a fixed binary-to-base encoding,
- Distributed cloud storage and fragmentation.

Wet-lab operations are not applicable in real-time cloud environments, and therefore, there are no costs for DNA synthesis, sequencing, reagents, or laboratory equipment. The computational cost is also limited to data encoding/decoding and fragmentation, which are comparable to typical data transformation operations. Integrating physical storage of DNA can be considered a possible extension of the framework to the archives, but it would primarily support long-term cold storage layers rather than cloud transactions sensitive to latency.

3.3 Cost considerations of DNA-based systems

The cost of biological methods of DNA storage is also quite high because DNA strands are expensive to produce, obtain, and amplify, and high-quality laboratory infrastructure is required. Current DNA synthesis technologies are slow and costly, and cannot be used in cloud services at high frequency or in real time. Moreover, sequencing and biochemical validation add latency and operational complexity.

Conversely, the suggested BioCrypt-DNA model will be based on a digital DNA abstraction model, in which the nucleotide bases (A, T, C, G) are treated as abstract characters encoded as binary data. Encoding, fragmentation, redundancy, and retrieval are all performed algorithmically in traditional CC setups.

Consequently, the cost-effectiveness boasted in this work is concerned with:

- Elastic cloud resource scalability,
- Effective systems of distributed storage and redundancy,
- Elimination of lab production and sequencing costs.

System deployment in biochemical synthesis requires no reagents or physical manipulation of DNA. The cost of operating BioCrypt-DNA will therefore be comparable to the standard encryption cost, plus the lightweight encoding overhead incurred in the cloud infrastructure.

3.4 DNA security and many attack types

To emphasise the significance of each attack against a DNA encryption system, this section summarises the attacks identified in the examined articles and their frequency in the research. The least complex and most significant assaults in the realm of networks are associated with this investigation (Kairi et al. 2023). Attacks are divided into both active and passive categories for network users. Active assaults are listed on a scale of 1 to 10, while passive and weak attacks are covered in the last part, as they are less commonly mentioned in the examined research.

The classification of various types and forms of cyber-attacks is elucidated in Fig. 1. Categorised cyber-attacks are network-based, application-based, social engineering, cloud-based, IoT-based, cryptographic, physical, wireless, and AI/ML-based attacks. Each of these categories entails specific attacks businesses may encounter. Network-based attacks include Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS), Man-in-the-Middle (MITM), Packet Sniffing, IP Spoofing, ARP Spoofing, etc. Application-based attacks include but aren't limited to SQL Injection, Cross-Site Scripting (XSS), Buffer Overflow, and Zero-Day Exploits. It also shows Social Engineering attacks like Phishing, Spear Phishing, and Vishing, as well as cloud-based attacks like Cloud Misconfiguration and Account Hijacking. It also considers IoT-based attacks such as Device Hijacking and Botnets, Brute Force Cryptographic attacks and Side-Channel Attacks, Physical attacks such as Hardware Tampering and Dumpster Diving, Wireless attacks such as Evil Twin and Bluejacking, AI/ML-Based Adversarial Attacks and Data Poisoning. This categorization can go a long way in understanding the multidimensionality of

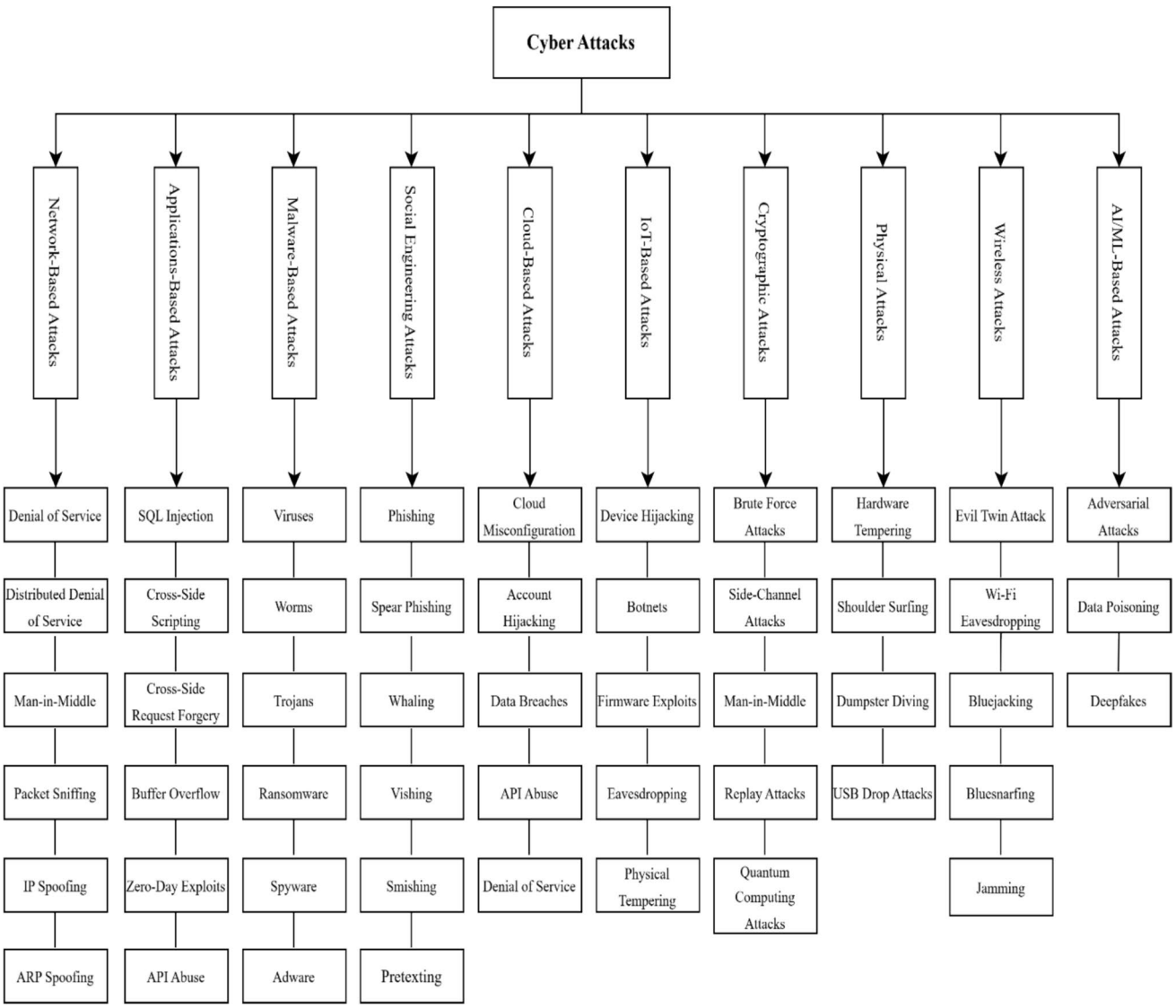


Fig. 1 A taxonomy of cyber attacks

cybercrimes and their various forms in various dimensions of the threat environment.

(a) Network-based attacks

Infrastructure attacks target network protocols or devices, including information theft and even access blocking (Zare et al. 2024). A DoS attack incapacitates a network's correct users by flooding it with traffic. The more advanced form is the Distributed DoS attack, which uses numerous systems to bombard a chosen target. These techniques are much more violent and focus far less on stealth than on using the victim's system as a sniper's nest. Man-in-the-middle, or "sniffer," attacks involve eavesdropping on two connected machines and attempting to extract information, such as passwords or financial details, from their communication without being directly involved.

In terms of obtaining information, Packet Sniffing focuses on monitoring and capturing data as it passes through the network, enabling the acquisition of logins and even credit card numbers. Using IP Spoofing, sending a packet to a network while pretending to be another device to attempt to erase existing security protocols is possible. ARP Spoofing helps an individual to transmit Interception messages to the next gadget. With the modification of ARP (Address Resolution Protocol), the computer performing the attack can manipulate the message so that it passes the data to be intercepted or for unauthorised access.

(b) Application-based attacks

Application attacks exploit vulnerabilities in software applications, web services, or APIs. One of them is SQL Injection, in which malicious SQL queries are injected into a web form to retrieve or remove sensitive data (Robert et al. 2024). Cross-site scripting (XSS) injects malicious scripts into web pages viewed by users to steal session cookies and redirect users to other websites. Cross-Site Request Forgery (CSRF) tricks the user into performing tasks in a web application where the user is already authenticated, such as modifying account details or transferring money. Buffer Overflow exploits flaws in memory allocation to overwrite memory addresses, often to execute code or crash the application. Zero-day exploits likewise target previously unknown flaws in specific pieces of software, such as bugs that developers have not yet discovered and cannot patch.

(c) Malware-based attacks

As the name implies, the second type of attack targets a specific malicious system, such as software or a program. This attack is exceptionally dangerous, as it can destroy, disrupt, or gain unauthorised access to confidential systems. Adware is a type of computer virus that causes significant problems for legitimate programs and documents, often moving deeper into systems and altering or stealing relevant data (Robert et al. 2024). Another type is Worms, which replicates itself across computer networks while consuming enormous bandwidth. Advertising malware, infamously known as adware, messes with computer systems by blasting unsolicited pop-up advertisements; it also hampers privacy by storing sensitive data to categorise users for advertising. Spyware stealthily tracks user activity, hoarding sensitive data such as passwords, usernames, and visited websites. Spyware, like most malware, often results in privacy violations.

(d) Social engineering attacks

Social engineering attacks exploit human psychology to extract sensitive information or coerce individuals into performing actions (Birthriya et al. 2024). Phishing involves sending deceitful emails or messages to recipients to obtain highly sensitive information, such as passwords or credit card numbers. Spear Phishing is a specific type of attack targeting individuals, businesses, or organisations that rely on contextually relevant details for greater success. Whaling is an advanced form of phishing done on high-value targets like directors or CEOs with detailed and sophisticated whaling attacks aimed at stealing sensitive corporate data. Vishing involves impersonating someone over the phone to deceive victims into disclosing sensitive data. Smishing is the SMS version of phishing, in which victims are tricked into clicking on malicious links or revealing sensitive information. Involve pretending to be a person or institution, like a bank or an IT support centre, with authority to fabricate a scenario to collect sensitive data, which is referred to as "Pretexting.

(e) Cloud-based attacks

Cloud attacks target cloud infrastructure, such as data, virtual machines, and services. An example is open public storage buckets, commonly referred to as cloud misconfiguration, which pose a significant threat to sensitive information (Mani 2024). Account Hijacking is the unauthorised access to cloud accounts through phishing or the use of stolen credentials. Data breaches occur when hackers gain access to sensitive information stored in a cloud database

due to weak security or inappropriate configuration. API Abuse concerns using cloud APIs without permission to gain access, to sabotage services, or to download data. Disruption of cloud service availability due to overload can be termed a Denial-of-Service (DoS) attack, targeting critical enterprise business operations.

(f) IoT-based attacks

Devices in the Internet of Things that are often poorly secured are frequently targets of IoT attacks. Taking control of IoT devices is called Device Hijacking and is done for illegal purposes, such as building bot networks (Yu et al. 2025). Many attacks are carried out through compromised IoT devices called Botnets; DDoS is an example of a large-scale attack. Gaining access to or collecting information from an IoT device can be achieved by exploiting its firmware, also known as Firmware Exploits. Sensitive information is obtained by intercepting data exchanges through eavesdropping. Physical Tampering involves altering the device to create boundaries and disrupt its normal functioning.

(g) Attack of cryptography

Cryptographic attacks attempt to compromise cryptographic systems to break the encryption or steal the necessary keys. Brute-force attacks attempt to crack encryption algorithms by trying every possible combination for a given encryption algorithm, focusing on weak keys and passwords (Ali and Anwer 2025). Side-channel attacks exploit physical emissions, such as power consumption and electromagnetic radiation, to recover cryptographic keys. In MITM (Man-in-the-Middle) Attacks, the hacker steals information or injects unwanted data by intercepting encrypted traffic and redirecting it through a malicious proxy. Replay Attacks impersonate users to capture and retransmit encrypted information, thereby gaining unauthorised access. Quantum Computing (QC) Attacks are those in which a quantum computer is used to defeat classical encryption algorithms such as RSA and even ECC.

(h) Attack in the physical layer

Physical attacks aim to gain direct access to the infrastructure or hardware. Hardware Tampering seeks to modify hardware to extract or access information, usually by making malicious changes (Liang et al. 2025). Shoulder Surfing involves closely watching users as they enter sensitive information, such as passwords or personal identification numbers, to acquire it. Dumpster Diving involves searching through waste for sensitive materials such as documents or hardware. USB Drop Attacks involve dropping USB flash drives in various public locations with the expectation that some users will connect them to their machines.

(i) Wireless attacks

A wireless assault is an attack aimed at wireless devices and networks. The Evil Twin Attack takes over a Wi-Fi network by setting up a parallel one to intercept traffic or capture login details. Wi-Fi Eavesdropping captures open Wi-Fi traffic to collect private information (Lee et al. 2025). Bluejacking sends unsolicited spam messages to Bluetooth-enabled handsets, with communication one-way, often for impersonation or harassment. Bluesnarfing involves extracting contacts or messages from a Bluetooth-enabled device. Communication Jamming modifies clear wireless signals to introduce noise and obscure them, often to bypass security measures.

(j) AI/ML-based attacks

AI/ML-based attacks use artificial intelligence or machine learning to increase the damage incurred. Adversary Attacks are targeted attacks against AI/ML systems in which the system's output is intentionally forced to depend on specific inputs, often to bypass security (Shahzad et al. 2025). Data poisoning is malicious data injection, first and foremost, such as injecting malicious data into a training set of an untrained AI/ML model to destroy it. Deepfakes are a new technology that uses AI to create spoofed audio, video, or images, often for social engineering crimes.

4 Cloud computing

Major companies like Google, Microsoft, and Amazon offer CC, a modern on-demand computing service. This service treats an application framework as a cloud, granting users and organizations global access. The fundamental principle of CC is to deliver computing power, storage, and software as a service. Despite varying definitions, CC generally shares several key features: pay-per-use (no upfront fees or utility charges), large-scale and scalable resource provisioning, a self-service interface, and resource isolation.

A cloud is a distributed, parallel system comprising networked, virtualized computers. These computers are dynamically deployed as unified computing resources, as specified by service-level agreements between customers and service providers (Ahmad et al. 2025; Mewada et al.

2025). According to the National Institute of Standards and Technology (NIST), CC is defined as a model that enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (such as servers, storage, networks, applications, and services). One can easily provision and release these resources with minimal management or engagement with the service provider. CC is like a data centre, delivered over the Internet. Customers will be charged based on their specific use of cloud storage, resources, and services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). This utility model provides processing power, storage, and resources, along with applications, to meet the diverse needs of customers. The cloud model possesses five fundamental characteristics, including resource pooling, on-demand self-service, high elasticity, broad network access, and measurable service. There are three primary service models: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). Also, there are four major deployment models: community, hybrid, private, and public clouds.

4.1 Challenges in cloud computing

Many firms and scientists are considering CC, which offers numerous advantages, but they face some challenges. Application security is one of the biggest obstacles, as cloud-based applications are vulnerable to injection attacks, API misconfiguration, and poor development practices. These weaknesses might lead to unauthorized access, breaches or service disruptions (Subramanian and Jeyaraj 2018). The other significant concern is data storage security, where breaches, losses, and non-adherence to data residency laws affect the masses stored. It is always a problem that confidentiality, integrity, and data availability will never be guaranteed (Mell and Grance 2011). Moreover, as dependence on cloud services has increased, network infrastructure has become a target of various attacks, including MITM, DDoS, and eavesdropping; thus, network security has increased in tandem with these services. They must use secure communication protocols and strong network monitoring to minimize these risks.

Client management is also very difficult, particularly regarding user access and permissions. Weak enforcement of authentication processes, privilege abuse, and malware infections of the cloud server infrastructure are multi-dimensional issues that may further sidetrack cloud security from core operational system (OS) concerns. Patching, antivirus defense, server hardening, and care should be performed frequently to ease the situation. (Mell and Grance 2011). Lastly, trust in CSPs is essential because users rely on them to manage and protect their data and ensure service availability. Trust is destroyed through service failures, vendor lock-in, and a lack of transparency. Companies need to closely evaluate whom they give these advanced technologies to and adopt multi-cloud strategies to reduce reliance on stagnant or low-quality suppliers. (Subramanian and Jeyaraj 2018; Mell and Grance 2011).

Compliance and legal constraints can also be listed as further considerations; these include regulatory specifications such as the GDPR or HIPAA, which can be a nightmare in the cloud (Mell and Grance 2011). Neither insider attacks (regardless of their intent) nor resource-sharing problems in multi-tenant clouds can be addressed without catastrophic consequences. A multi-faceted approach should be adopted to overcome these challenges, including strong encryption, robust access controls, regular auditing, and compliance with security policies and practices. The identification and the absence of these challenges allow better protection of CC and enhance the efficiency of organizations.

The detailed overview of data security in CC, presented in Fig. 2, is split into several issues. Data loss, breaches of privacy, and violations of data privacy are key issues associated with poor storage and ineffective data encryption. Network Security issues involve man-in-the-middle, DDoS, DNS, and Insecure API attacks, as they allow unauthorized access to accounts and service interruptions. Weak authentication, credential theft, abuse of privileges, and insider threats are among the identity and access management (IAM) issues that compromise system integrity. The areas of compliance and legal concern include flagrant violations of laws and regulations, data sovereignty, jurisdictional issues, and compliance lapses by cloud providers serving global businesses. The problems of virtualization and multi-tenancy are also very important. These risks include VM breaches, side-channel infiltration, broken tenant surrogacy, and resource exhaustion. These vulnerabilities allow hackers to exploit shared cloud resources. Cloud service provider risks include vendor lock-in, limited service, system downtime, and malfunctioning security systems, which result in breaches of trust and make it difficult to define responsibilities between providers and users. Application security vulnerabilities are caused by insecure coding practices, injection attacks, malware, ransomware, and poor patch management, which facilitate exploitation. There are some people-related aspects: insider abuse, social engineering attacks, low-security culture, and carelessness also result in breaches. Finally, disaster recovery and business continuity are related to ineffective backup policies, prolonged outages from natural disasters, and insufficient disaster response plans, which further exacerbate the effects of critical security incidents.

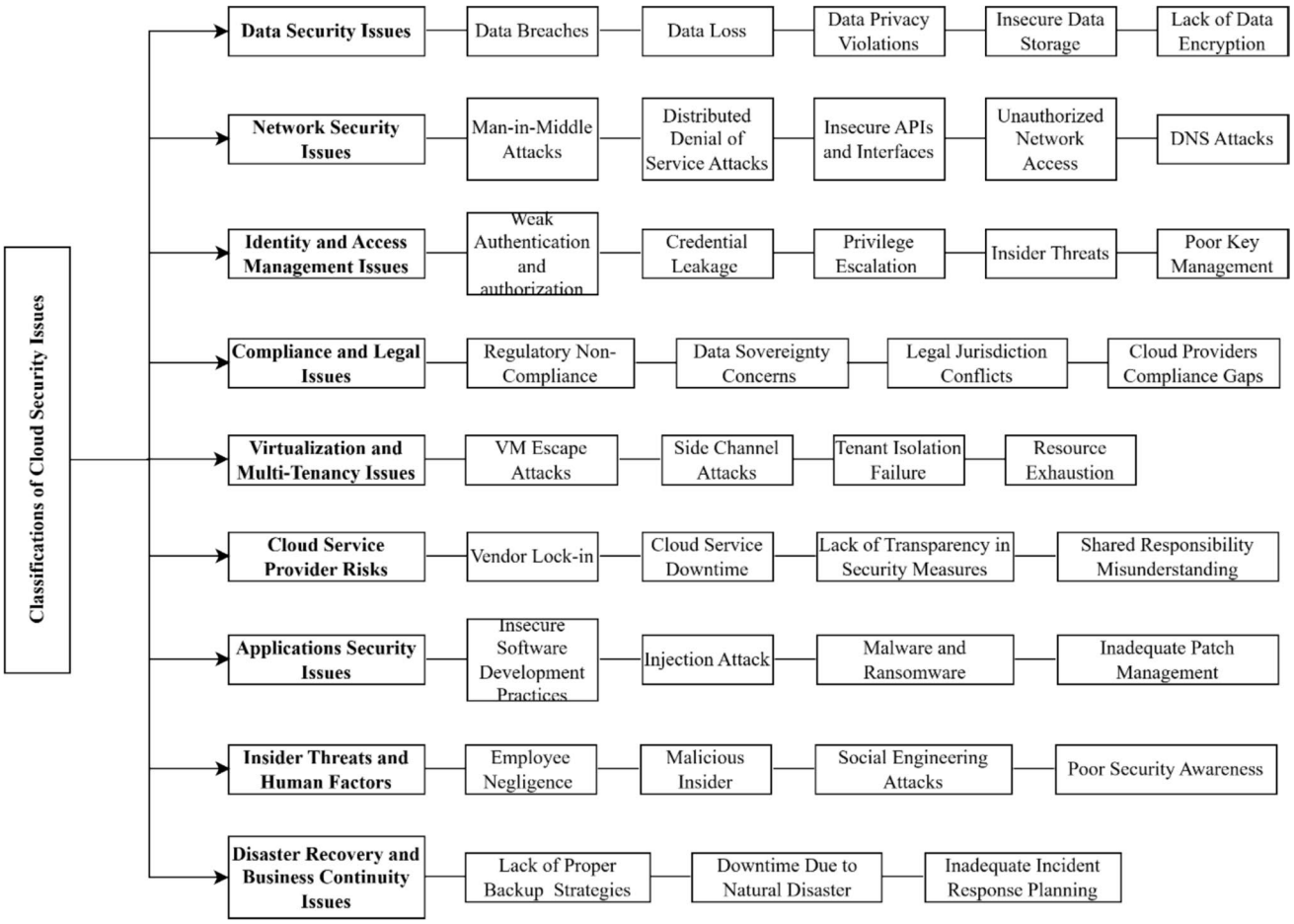


Fig. 2 Classifications of cloud security issues

4.2 Importance of cloud data security

Cloud data security has become a very important issue in the wake of the increased migration of devices, data centres and business functions to the cloud. This has been influenced by the need to protect sensitive information and ensure the integrity of the cloud-based systems. Strong cloud information security relies on innovative security technologies, a strong corporate security culture, and proactive security policies. Cloud security, or CC security, refers to the policies, procedures, and technological controls that protect cloud-based systems, data, and infrastructure. These policies play a vital role in protecting customer privacy, ensuring compliance, ensuring cloud data security, and meeting user and device authentication requirements. The targeted access to business challenges and proactively hone them can be customized by organizations based on traffic filtering to reduce access to business challenges. In response to the Elastica Q2 2015 initiative and similar programs, the Cloud Security Alliance (CSA) is moving toward creating effective, risk-reducing cloud application security frameworks. These frameworks can assist organizations in protecting their cloud environments against the ever-changing threats.

Moreover, the International Organization of Standardization (ISO) offers the basic data security standards that are necessary to introduce more sophisticated technical solutions. These standards will also ensure that the security configurations of the cloud are set to international standards to allow organizations to deploy secure cloud architectures. The collaboration among industry leaders, standardization organizations, and other institutions will be of interest for overcoming new security challenges and supporting the safe and effective use of cloud technologies as their use grows.

4.3 Cloud computing security requirements

As a CSP and third-party service provider to a company, you have robust security and safety measures in place to protect the information you handle. According to NIST (Mauritsius and Saputra 2019), data confidentiality, integrity, and availability should be central to an Information Technology (IT) system. CSPs should not disclose sensitive information to anyone but authorized personnel, as this will help prevent

Table 2 Cloud Security: Needs and Challenges

Area	Specific aspect	Explanation
Essential Security Needs	Data Privacy	It protects sensitive client information and restricts access to authorised individuals only
	Data Accuracy	Maintains data integrity, ensuring it remains unchanged and correct during processing or transfer while allowing authorised modifications
	Service Accessibility	It guarantees the consistent availability of services and information accessible anytime, anywhere
	User Verification	Confirms user identities to grant service access and safeguard account security
	Access Control	Limits access rights to users who provide valid credentials
Common Cloud Security Issues	Incorrect Setup	Improper configurations are a major vulnerability, potentially exposing data and applications. Strong configurations, encryption, access controls, and robust passwords are vital
	Unauthorised Entry	Unpermitted access to data is a frequent problem. Businesses must limit access to authorised personnel and use encryption and backups to protect data
	Account Takeover	Account hijacking remains an important concern. Users should protect their accounts by creating strong passwords, setting up security questions, enabling two-factor authentication, and regularly monitoring account activity
	Limited Oversight	The security of cloud servers is a worrying matter because data is accessed from remote locations. Companies ought to have strong authentication, loss-prevention, and intrusion-detection systems. Regular security reviews are also important
	Privacy Concerns	CC's global accessibility raises privacy issues. Companies must ensure that only authorised users access data and implement security measures to prevent breaches
	Third-Party Data Sharing Risks	Working with external providers increases the risk of data loss or unauthorised access. Strong security measures, such as encryption and stringent internal data management policies, are necessary
	Compliance Requirements	CC introduces regulatory challenges. Organisations must secure data and adhere to legal standards through regular audits, strong authentication, and software updates
	Risks from External Sources	Third-party resources outside the cloud provider's control present security threats. Businesses should rely on trusted sources and enforce strong security controls, such as multi-factor authentication

breaches and interference. It is also important that the data be accurate throughout processing or transmission. The user must be able to access, change, or delete data without altering its structure. Besides, CSPs are expected to ensure the reliability of their services and systems, ensuring availability at any time to enhance convenience and trust, while maintaining regulatory compliance. In addition, identity authentication ensures that the identity users claim matches who they are before they can access services. Only users with the correct credentials should be allowed to access the privileges. This interaction establishes a state of trust, particularly in the cloud-centric world that is increasingly reliant on it.

The basic security needs of CC, as pointed out in Table 2, are confidentiality, integrity, availability, authentication, and authorization. Moreover, it describes the typical security issues faced by cloud users. Technical issues, such as misconfigurations and unauthorized access, invasion of privacy, sharing data with third parties, legal compliance, and dependence on obscure external resources, are all types of security problems. The table reveals that to ensure cloud systems run securely, it is necessary to implement strong security measures, including but not limited to strong authentication, encryption, periodic security audits, and access controls.

5 Proposed methodology

The paper introduces a new way of thinking about cloud data security, focusing on the proposed methodology. BioCrypt-DNA is a hybrid that incorporates DNA-based encryption, steganography and indexing to safeguard sensitive information, and it is scalable and efficient. BioCrypt-DNA differs from conventional DNA cryptography in that it does not operate on physical DNA sequences; rather, it uses digital DNA to break down documents, ensuring maximum flexibility and processing power. The approach relies on four nucleic acid bases A, C, G and T, with A complementing T, G complementing C, and G and T complementing G. Although the principle of encoding based on DNA introduces additional computational steps to the classical encryption, these are lightweight symbolic mapping and fragmentation procedures to be performed in pure software. It is therefore not biochemical but computational and can be run in real time and in the cloud. Scalability and efficiency, as defined in this framework, imply elastic computing, distributed storage, and parallel processing. BioCrypt-DNA will not require strand synthesis or sequencing, unlike biological systems for DNA storage, which can create bottlenecks in synthesis speed and in laboratory capacity. The DNA-based encryption in this item involves digitally simulated encoding of DNA, as well as biochemical manipulation of

DNA strands. Encoding, fragmentation, and reconstruction are implemented algorithmically in the cloud infrastructure. This design alternative aligns real-time cloud environments and circumvents the resource constraints (financial and operational) of DNA synthesis and sequencing technologies in the laboratory. The following lines outline the key components of the suggested framework.

- Encryption of data

Text files, pictures, or sensitive documents are encrypted using standard cryptographic algorithms such as AES or RSA before being converted to a secure format, which is then encoded in DNA.

- Encoding through DNA

The encrypted data, in binary bits or characters, is encoded into a coding scheme that forms the basis of the DNA sequence. The conversion of data into a given DNA sequence is achieved by mapping binary bits or characters to the nucleotide bases A, T, C, and G in a specific sequence. This makes it safer as biology introduces randomness to cryptography.

- Fragmenting and distribution in the cloud

The encoded section of the DNA is fragmented into a few smaller pieces to enhance storage efficiency and redundancy. The fragments are then reliably deployed to multiple cloud storage systems to enhance data availability to withstand faults and failures.

- Cloud storage and redundancy

The encrypted, fragmented DNA sequences are then stored within the cloud services. The fragments undergo redundancy techniques to protect against loss or corruption and ensure data durability.

- Secure access control and key management

Robust access control mechanisms and secure key management systems within BioCrypt-DNA are integrated to safeguard the DNA-encoded information and restrict unauthorised access. This ensures that only specific users can retrieve and decode the protected information.

- Data retrieval and DNA decoding

After DNA-encoded fragments are retrieved, the information is reassembled and decoded back to the original encrypted digital format. This uses a reverse DNA encoding scheme to ensure data accuracy and consistency.

- Decryption process

Encrypted information is decrypted using the appropriate cryptographic key and algorithm for the data (e.g., AES or RSA) to recover the original plaintext securely.

- Data integrity verification

Hash values and the checksum mechanism verify decrypted data, ensuring no unauthorised changes while the data is being stored or retrieved.

- Deletion and modification using security measures

To comply with regulations and data privacy requirements, BioCrypt-DNA uses secure deletion and update mechanisms that permanently delete sensitive data and data that are no longer relevant, without any possibility of recovery.

A decade ago, BioCrypt-DNA created a cloud storage solution by utilizing DNA cryptography that is simultaneously highly secure, scalable, and efficient. By combining biological methods of encryption with cloud storage security, BioCrypt-DNA increases the confidentiality, integrity, and availability of the data, thereby providing the framework for the next generation of cloud security solutions BioCrypt-DNA is known for.

Figure 3 depicts a basic data security flow that includes the first two steps: data encryption and data encoding. The flow adds steps to securely delete or update data after encoding and encryption are complete, so sensitive data can be removed or modified when necessary. This workflow exemplifies how data security is paramount, from encryption to the final disposal or information change.

Table 3 illustrates mapping the four bases of DNA, A, G, C and T, to their binary 4-bit representations. Specifically, A equals 0000, G equals 0001, C equals 0010, and T equals 0011. This allows DNA strands to be represented in binary form, such as DNA encryption, encoding, or even other operations in biocomputing and cryptography. The table is a basic example of how biological information can be transformed into a digital format for processing and analysis.

The BioCrypt: DNA-Based Cloud Security encryption algorithm will first encrypt input plaintext P with a cryptographic algorithm (e.g., AES, RSA) and encryption key K to generate encrypted data E. Then E is encrypted and transformed into a DNA sequence S with the help of a predetermined encoding scheme, which links digital symbols to DNA bases {A, T, C, G}. The DNA sequence S is then divided into smaller information units $\{F_1, F_2, \ldots, F_n\}$, each

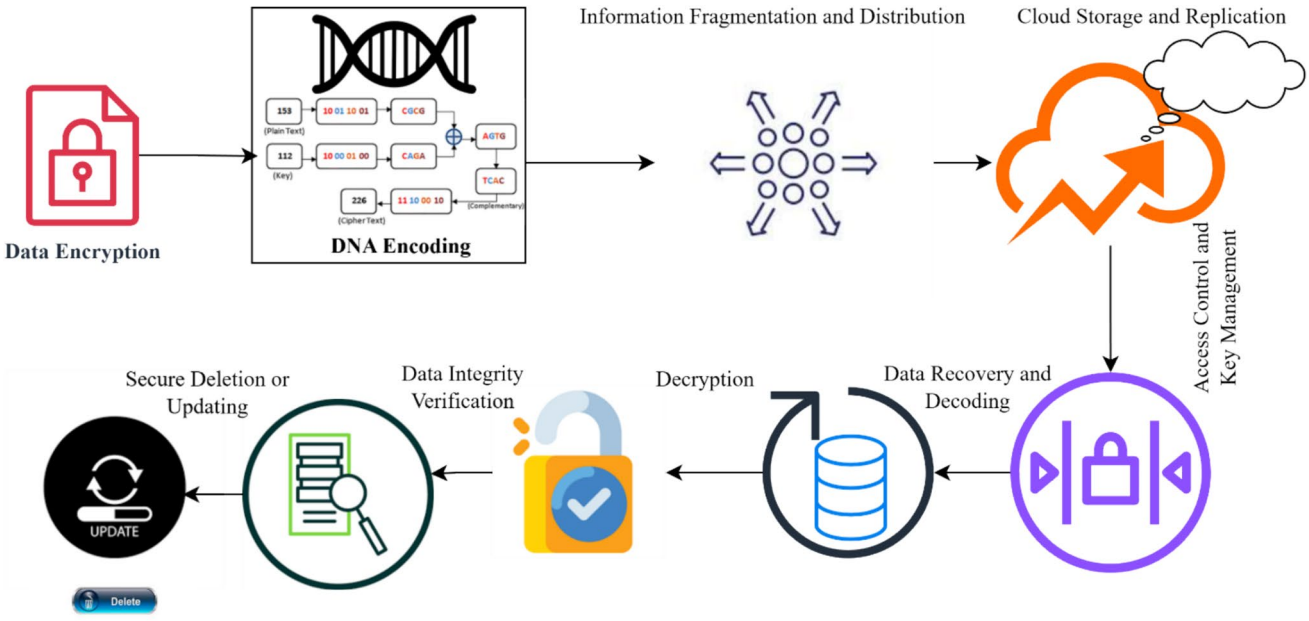


Fig. 3 Proposed methodology

Algorithm 1 BioCrypt: DNA-based cloud security encryption

Input:
P: Plaintext (digital data to be encrypted).
K: Encryption key.
Output:
F: Fragmented DNA-encoded data stored in the cloud.

Steps:

1. Digital Encryption:

- Encrypt the plaintext P using a cryptographic algorithm (e.g., AES, RSA) with the encryption key K.
- Let *E* represent the encrypted digital data:

$$E=\text{Encrypt}(P, K)$$

2. DNA Encoding:

- Define a mapping table M that associates digital symbols (e.g., bits or bytes) with DNA bases {A, T, C, G}.
- Map the encrypted data E into a DNA sequence S*S* using the mapping table M:

$$S=\text{Map}(E, M)$$

Here, S is a sequence of DNA bases {A, T, C, G}.

3. Fragmentation and Concatenation:

- Divide the DNA sequence *S* into smaller fragments $\{F_1, F_2, \ldots, F_n\}$, where each fragment F_i is fixed or variable.
- Concatenate metadata M_i (e.g., position, fragment ID) to each fragment F_i:

$$F_i' = F_i \parallel M_i$$

Compute checksum or hash Hi for each fragment Fi′ to enable integrity verification during retrieval.

- The set of all fragments with metadata is:

$$F= \{F_1', F_2', \ldots, F_n'\}$$

4. Apply Forward Error Correction:
Generate r parity fragments using Reed–Solomon RS(n, k) coding and append them to the fragment set before cloud storage.

5. Cloud Storage and Replication:

- Store the fragmented DNA-encoded data *F* in the cloud storage infrastructure.
- Replicate the fragments across multiple cloud servers for redundancy and durability:

$$\text{Store}(F, \text{Cloud})$$
$$\text{Replicate}(F, \text{Cloud})$$

Table 3 Binary table with DNA bases

DNA base	Binary value
A	0000
G	0001
C	0010
T	0011

accompanied by metadata Mi (e.g., position, fragment ID), which makes the fragments easier to identify and reassemble. The fragmented DNA-encoded data FF is then stored, replicated, and distributed across the cloud-based infrastructure to ensure durability and availability. The algorithm provides a reliable, effective solution for encrypting and storing information in the cloud using DNA-based cryptography.

The BioCrypt: DNA-Based Cloud Security Decryption algorithm starts by retrieving DNA-encoded data fragments F from cloud storage, i.e., F = {F1', F2', …, F_n'}. The next step involves removing metadata Mi from each fragment Fi, followed by concatenating the DNA-encoded segments {F_1, F_2,…, F_n} to form the complete DNA sequence. The vital step follows, in which the full DNA sequence SS is decoded back into the encrypted digital data EE using the inverse of the previously specified encoding scheme, M-1. The digital data E is decrypted using the decryption key K to obtain the original plaintext PP. This algorithm ensures accurate, secure reconstruction and decryption of data stored in the cloud, leveraging DNA-based cryptography.

5.1 Fragmentation and distribution in the cloud

The DNA-encoded sequence is broken into tiny fragments to enhance storage flexibility, parallel processing and fault tolerance. Individual fragments are terminated with structured metadata, including the fragment ID, sequence order, checksum value, and optional parity symbols.

In order to guarantee sound reconstruction, the BioCrypt-DNA will have multi-layer redundancy:

- The fragments are copied onto several geographically spread cloud nodes to provide availability in the event of the failure of a cloud node.
- Several optional forward error-correction encodings, like Reed-Solomon, could be used to produce parity fragments. This will enable the recovery of lost fragments without necessarily making complete copies.
- Each fragment is stored by calculating cryptographic hash functions (e.g. SHA-based digests) before storage. Hash verification is useful during retrieval to get access to tampering or corruption.

Because BioCrypt-DNA is digital, it does not suffer from biochemical noise (e.g., strand degradation or sequencing errors). Nevertheless, the framework leverages robust cloud storage reliability mechanisms to mitigate corruption, packet loss, and bit errors at the storage level. In reconstruction, metadata and integrity are used to validate the fragments, and then the fragments are concatenated. In the event of fragment loss, redundancy can be used to recover without affecting data accuracy.

Algorithm 2 BioCrypt: DNA-based cloud security decryption

Input:
F={F_1', F_2',…, Fn′}F={F_1',F_2',…,F_n'}: DNA-encoded data fragments retrieved from cloud storage.
K′: Encrypted decryption key (retrieved from secure storage).

Output:
P: Decrypted plaintext (original digital data).

Steps:

Retrieve Data:

- Retrieve the DNA-encoded data fragments F= {F_1', F_2',…, Fn′}F={F_1',F_2',…,F_n'} from the cloud storage:

$$F=\text{Retrieve (Cloud)}$$

Concatenation and Reconstruction:

- Remove metadata M_i from each fragment F_i' to obtain the DNA-encoded segments F_i:

$$F_i = \text{RemoveMetadata}(F_i')$$

- Concatenate the DNA-encoded segments {F_1, F_2,…,F_n}to reconstruct the complete DNA sequence SS:

$$S=F_1 \| F_2 \| \cdots \| F_n$$

DNA-to-Digital Decoding:

- Decode the DNA sequence S back into encrypted digital data E using the reverse of the defined encoding scheme M^{-1}:

$$E=\text{Decode}(S, M^{-1})$$

Digital Decryption:

- Decrypt the encrypted digital data EE using the decryption key K (obtained by decrypting K′):

$$P=\text{Decrypt}(E, K)$$

5.2 Error-correction using reed–solomon coding

To ensure that DNA-encoded fragments of the data can be reliably recreated from distributed cloud storage, BioCrypt-DNA provides an optional forward error correction (FEC) mechanism based on Reed-Solomon (RS) coding. Reed-Solomon codes are a commonly used technique in distributed storage systems because they can reconstruct lost or damaged fragments without replicating all of them.

The error-correction process operates as follows:

Step 1: DNA Encoding Completion

The encrypted data is next mapped to a DNA sequence, broken into k primary fragments, and encrypted.

Step 2: Symbol Conversion for Coding

All DNA fragments are transformed into finite-field symbols that can be processed by Reed-Solomon (e.g., GF (28)).

Step 3: Parity Generation

In RS (n, k) coding, r = n -k parity fragments are obtained. These pieces of parity information are mathematically generated redundancy information.

Step 4: Distributed Storage

The overall number of fragments (k original + r parity) is spread across autonomous cloud storage nodes.

Step 5: Fragment Retrieval and Validation

In retrieval, fragments are gathered in the available nodes. If r fragments are lost or corrupted, the RS decoder will reconstruct the original k fragments using syndrome-based error detection and correction.

Step 6: Reconstruction and Decoding

After the fragments are found, metadata indexing is used to concatenate them in the correct sequence, and they are then decoded to produce encrypted digital data.

Because BioCrypt-DNA is digital, storage node failures, transmission errors, and data corruption are mitigated by Reed-Solomon error correction rather than biochemical mutations. Such layered redundancy can effectively enhance reliability without compromising storage efficiency.

6 Results and discussion

The given performance evaluation is based solely on computational costs related to encryption, symbolic encoding of DNA, fragmentation, and redundancy operations. There is no physical DNA synthesis or sequencing involved in analysis or implementation. The performance analysis shown in this section focuses solely on purely computational operations, such as encryption, DNA-symbolic encoding, fragmentation, and distributed storage simulation. There are no biochemical DNA synthesis and sequencing procedures in the experimental setup. Thus, the mentioned efficiency in terms of time is related to software implementation in the cloud-compatible infrastructure. The assessment of the obfuscated encryption method, which is claimed to be implemented using the DNA framework, was conducted with several important aspects in mind. These include requirements for efficient cryptographic performance, key space, key and plaintext sensitivity, size comparisons, and time complexity. Strong security was achieved by ensuring that the data is protected by both the key and the input data. Therefore, it is impossible to attack where the deductive reasoning of the key from the cypher is in use. The analysis of the key space revealed a security strength of 8000 bits, which is indeed strong but remains vulnerable to brute-force attacks. However, the method exhibited high key and plaintext sensitivity, with a bit change percentage of 85.25%, significantly higher than the recommended 50%, making it difficult for attackers to decipher the ciphertext by analysing plaintext or key attributes.

Table 4 compares the performance of various DNA-based cryptographic algorithms, including DNA-One Time Password (OTP), Time Password, DNA-RSA, standard RSA, Compressed DNA-RSA, and the proposed algorithm, across different plaintext sizes. It displays the relationship between the original file size in characters, the original text size in bytes, and the ciphertext size in bytes. It is important to note that DNA-RSA and standard RSA generate ciphertexts that are much larger than those of the OTP and the proposed algorithms. That compressed DNA-RSA produces ciphertexts that are smaller than the uncompressed ones. The proposed algorithm tends to produce ciphertexts of intermediate size, suggesting a trade-off between security and data overload.

Table 5 shows the performance of the proposed algorithm, including encryption and decryption times in milliseconds, for plaintexts of varying sizes in bytes. The time

Table 4 Performance of various DNA-based cryptographic algorithms

Algorithms	Original file in char	Original text in bytes	Cipher text in bytes
DNA-One Time Password (OTP)	550	1070	4352
	1550	1685	6912
	2550	3070	12,288
	3550	4834	19,456
	4550	6056	24,320
OTP	550	1070	1280
	1550	1685	1792
	2550	3070	3072
	3550	4834	4864
	4550	6056	6144
DNA- Rivest-Shamir-Adleman (RSA)	550	1070	113,028
	1550	1685	177,868
	2550	3070	324,324
	3550	4834	510,320
	4550	6056	651,376
RSA	550	1070	32,694
	1550	1685	39,836
	2550	3070	54,793
	3550	4834	86,284
	4550	6056	103,945
Compressed DNA-Rivest- Shamir-Adleman (RSA)	550	1070	40,164
	1550	1685	57,872
	2550	3070	91,948
	3550	4834	110,828
	4550	6056	106,632
Proposed Algorithm	550	1080	6055
	1550	1688	12,087
	2550	3071	24,168
	3550	4837	36,255
	4550	6057	48,339

Table 5 Time comparison of the proposed algorithm

Plaintext size (bytes)	Encryption time (ms)	Decryption time (ms)
10	10.351	10.727
20	10.707	10.715
40	11.427	10.471
80	11.783	10.459
100	11.904	10.243
500	12.668	10.125
1000	13.212	10.123
2000	13.541	10.121
3000	15.925	10.11

of encryption tends to increase as the size of the plaintext grows, from 10.351 ms to 15.925 ms for 10 and 3000 bytes, respectively, which shows that the volume of data and the time of encryption are directly related. In contrast, the decryption time shows a steadier trend, remaining around 10 ms across all plaintext sizes, indicating that the decryption speed is less dependent on data volume than the encryption speed.

Figure 4 shows the encryption and decryption times (in milliseconds) for different plaintext sizes (in bytes). Encryption time varies from 10.351 ms to encrypt a 10-byte plaintext to 15.925 ms to encrypt a 3000-byte plaintext, and it increases with plaintext size. On the contrary, the decryption times are rather consistent, ranging from 10.110 ms to 10.727 ms, independent of the plaintext size. This shows that decryption is very efficient and predictable, and that the time required for encryption is proportional to the size of the data. The figure highlights the system's ability to provide encryption times that are manageable for working with larger datasets and to ensure decryption is always fast, making it suitable for applications where security can be demanded and fast decryption is required.

6.1 Role of DNA-inspired sequencing in data reconstruction and security

The DNA sequencing in BioCrypt-DNA does not relate to biological laboratory procedures. Rather, it is the algorithmic reassembling of digitally encoded DNA sequences of disseminated fragments. Data is reconstructed and divided into fragments after encryption and DNA mapping and spread across numerous cloud nodes. In retrieval, retrieved fragments are joined, checked for integrity using an integrity hash, reassembled using metadata-based indexing, and decrypted back to binary data.

This reconstruction with sequencing improves three aspects of security: (i) fragmented storage makes the entire data vulnerable to an attack of one node, (ii) controlled reassembling implies that the correct metadata and key authorization are necessary, and (iii) cryptographic decryption is appropriate to make even the correctly ordered sequences incomprehensible without appropriate keys. In this way, BioCrypt-DNA sequencing enhances the structural security and access control without biochemical interference.

6.2 Comparative analysis with standard cryptographic algorithms

To put the presented BioCrypt-DNA framework's performance into perspective, an overview of the cryptographic standards used in the market, including AES-256 and RSA-2048, is provided. AES-256 provides the key space of 2 136; it is generally considered to be computationally secure against brute force attempts, and RSA-2048 is a strong asymmetric security measure on the hardness of integer factorizations. It should be noted that BioCrypt-DNA is not an alternative to AES or RSA; it uses classical encryption (e.g., AES or RSA) as the foundational cryptographic layer and provides a DNA-based encoding and fragmentation layer to enhance structural and distributed security. Regarding

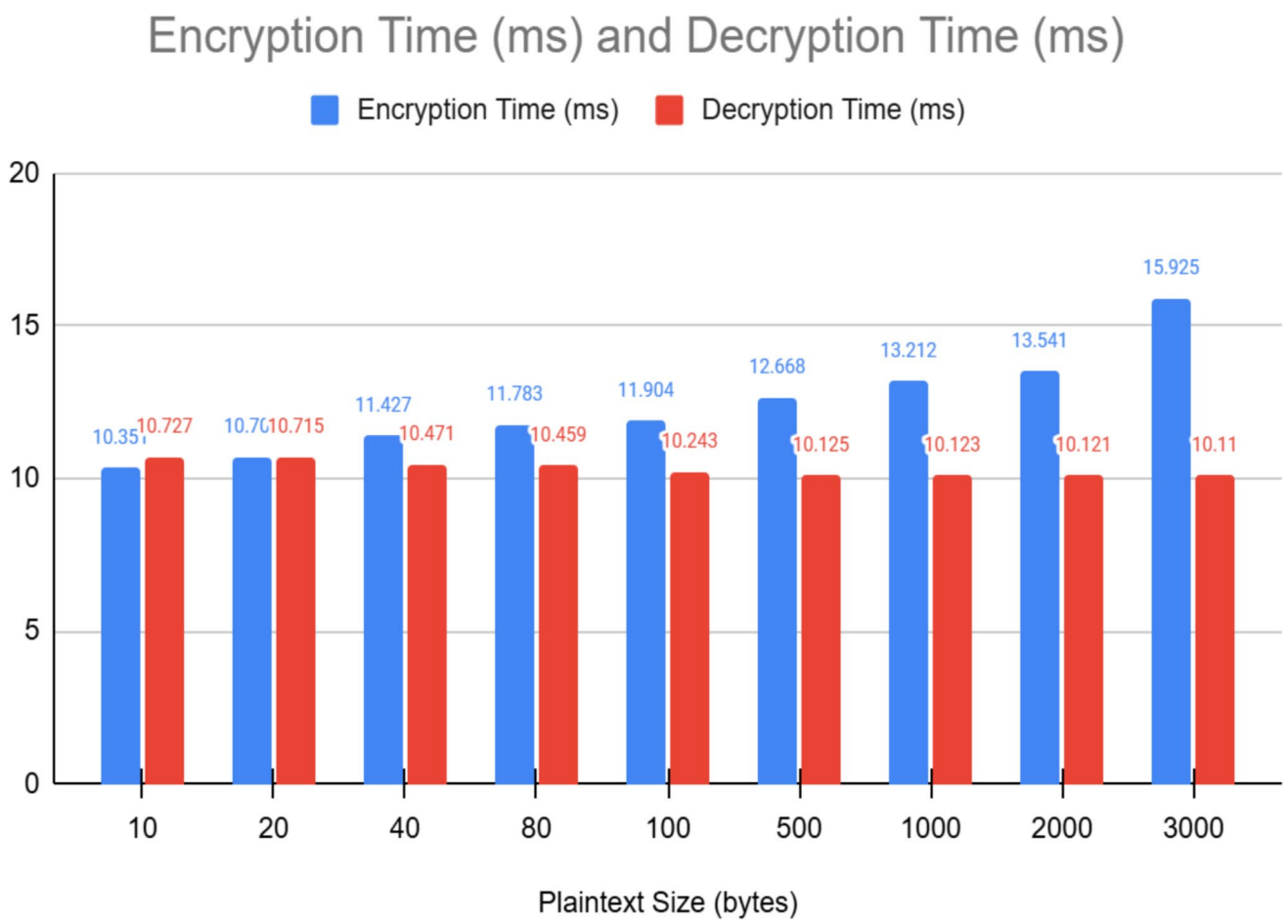


Fig. 4 Time comparison of the proposed algorithm

Table 6 Comparative overview with AES-256 and RSA-2048

Parameter	AES-256	RSA-2048	Proposed BioCrypt
Key Space	2256	~112-bit equivalent security	Depends on underlying AES/RSA + structural complexity
Encryption Speed	Very Fast	Slower	Slightly higher than AES due to encoding overhead
Decryption Speed	Very Fast	Slower	Comparable to AES
Structural Fragmentation	No	No	Yes
Distributed Cloud Redundancy	No (native)	No (native)	Yes
Resistance to Brute Force	Very High	Very High	Very High
Resistance to Reconstruction Attack	Moderate	Moderate	High (due to fragmentation + encoding)

key space, in BioCrypt, the effective cryptographic strength does not drop below 256 bits at the encryption level when AES-256 is employed. The additional DNA encoding does not diminish this strength but rather adds complexity to the structure and enhances resistance to data reconstruction attacks in distributed clouds. In terms of computational speed, the encryption and decryption times of BioCrypt-DNA are comparable to those of typical AES-based systems, and the overhead associated with DNA encoding, fragmentation, and metadata processing is moderate. Nevertheless, this is an overhead that cloud-based applications can accept. Compared with RSA-2048, which generally incurs higher computational costs for encryption and decryption, BioCrypt, coupled with symmetric encryption, offers better time efficiency for large-scale data and provides the same or greater security features.

Table 6 presents a comparative analysis of BioCrypt's ability to match the cryptographic integrity of AES-256 and RSA-2048, and to introduce novel structural fragmentation and native cloud redundancy. This design trades a marginal performance blow to achieve much greater resilience to

Table 7 Comparative encryption and decryption time analysis

Plaintext size	AES-256 (ms)	RSA-2048 (ms)	BioCrypt-DNA (ms)
Small (100B)	Very Low	Moderate	Slightly higher than AES
Medium (1 KB)	Low	High	Slightly higher than AES
Large (3 KB+)	Low	Very High	Moderate

reconstruction attacks, creating a more resilient security model of decentralized and distributed clouds.

6.3 Comparative time analysis with traditional encryption methods

A time analysis was used to analyze the computational efficiency of the proposed BioCrypt-DNA framework and compare it with the traditional encryption schemes, such as AES-256 and RSA-2048. AES-256 is widely used as a high-speed symmetric encryption algorithm, whereas RSA-2048 is an asymmetric algorithm that typically incurs high computational overhead, particularly when large amounts of data are involved. The findings suggest that BioCrypt-DNA adds an average time to standalone AES encryption, since it requires extra encoding steps on the DNA, metadata addition, and fragmentation. Nevertheless, this overhead is at the fringe and proportional to the input size. The decryption times are also like those of the AES-based ones, since the reverse-DNA decryption algorithm is computationally simple. BioCrypt, when used in the plaintext approach with symmetric encryption as the underlying cryptographic layer, shows much lower execution time for large plaintext sizes than RSA-2048. Thus, although DNA-inspired encoding provides low latency only for AES, the framework is computationally feasible in the cloud and significantly more efficient than entirely asymmetric solutions for bulk data processing.

Table 7 shows that the proposed BioCrypt-DNA system is computationally more efficient than older asymmetric standards. Although RSA-2048 has a weakness of scalability, i.e. with an increasing data size, its processing time is shown to have a high profile of "Very High", which, in contrast to that of AES-256, BioCrypt-DNA has a performance profile similar to that of AES-256.

7 Security analysis

Conventional DNA storage systems must deal with biochemical noise, such as mutations and sequencing errors, but BioCrypt-DNA is a fully digital system. Thus, its reliability issues are related to distributed storage faults, transmission errors, or node failures rather than biochemical instability. These threats are mitigated through replication schemes, erasure coding (e.g., Reed-Solomon), cryptographic integrity hashes, and organised fragment indexing, enabling correct, deterministic recovery. Although cryptographic tools ensure protection of confidentiality and controlled access, current cloud security also involves protection against behavioural and zero-day attacks. Further optimizing BioCrypt-DNA is based on implementing adaptive AI-based monitoring that would allow the construction of continuous anomaly detection that would complement its cryptographic and storage-level protection. The security analysis of the proposed system focuses on mitigating integrity and confidentiality threats to data stored in cloud environments. The approach has multiple elements that facilitate enhancing the security of information processes. Some knowledge is expected to be possessed by users through random sequences, a pre-coding and non-coding region, coding index values, some rules of binary coding, key values, and DNA sequences, all of which add to the complexity of the process. Apart from knowing all the elements that constitute the program, there are severe restrictions on access to the arbitrarily selected DNA sequences; these ensure that the program is highly complex, making it easier for those who need it to access it. The randomness of DNA selection adds to the complexity, as one has to select from a wide range, making it almost impossible to know which specific encryption sequence is chosen. The obfuscation, along with the intricate nature of finding the encryption key, provides strong security, minimising the risk of a data breach. Furthermore, the system addresses the dangers posed by CSPs, as aforementioned, by ensuring no data loss, calculated as Risk = P (data loss) * (the quantity of data loss). The proposed system also mitigates the risks associated with traditional services, thereby enhancing data security and protecting the confidentiality and integrity of information to sustain the modern IT business environment. Besides cryptographic protection, smart monitoring devices can also provide additional protection for the BioCrypt-DNA framework. Cryptographically, BioCrypt-DNA is based on the security of its underlying encryption methods (e.g., AES-256 or RSA-2048) and thus complies with existing cryptographic standards. In addition to classical encryption, the framework provides structural and storage-level security based on DNA-inspired encoding, controlled fragmentation, distributed storage, redundancy, and Reed-Solomon error correction. Deterministic reconstruction of fragmented data is also possible when Reed-Solomon coding is incorporated, even when a portion of the storage nodes fail, in which case the data remains available and intact, and cryptography maintains a high level of confidentiality.

Anomaly detection powered by AI can be further integrated into BioCrypt's cloud layer (Ahmad et al. 2025, 2026)

to enhance resilience. The latest studies on AI-assisted IoT security reveal that hybrid deep learning models (Kumar et al. 2025) are effective at identifying faulty behavior and zero-day attacks, which are difficult to detect with traditional rule-based systems. The combination of convolutional and recurrent neural networks can be developed to improve the ability to identify dynamic traffic patterns. By contrast, Generative Adversarial Networks (GANs) can mitigate class imbalance by training them to generate realistic samples of minority attacks. With the addition of such intelligent surveillance to BioCrypt, it will be possible to conduct behavioral analysis, recognize previously suspicious access attempts, and detect abnormal fragment access patterns and coordinated attack activities. This multi-layered approach brings in a vigorous responsive defence against known and emerging threats.

8 Conclusion

The new and viable model of big data security in modern cloud systems is BioCrypt, a platform that integrates cryptography, biometrics, and DNA-inspired symbolic coding principles. The structure improves confidentiality and structural protection by mapping encrypted data to the bases of DNA (A, T, C, and G), breaking the data down, and spreading the fragments across the nodes of the cloud. Redundancy, key control and optional error-correction mechanisms give recovery, reliable reconstruction, integrity and only authorized users access control. BioCrypt-DNA is completely digital and not constrained by the costly, delay-prone nature of biological DNA storage and stylus synthesis, as it operates entirely on a digital computational platform. Using a digitally simulated DNA-inspired encoding model eliminates biochemical costs and reliance on the laboratory, thereby ensuring real-time viability and affordability in cloud-based systems. Even though the extra encoding and fragmentation procedures introduce moderate computational overhead to the standalone symmetric encryption, a comparative analysis with AES-256 and RSA-2048 confirms that BioCrypt-DNA has standard cryptographic strength and the additional structural robustness of distributed storage systems.

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Declarations

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